

Explainable AI with Conditional Retrieval-Augmented Generation and Research Agent

Abstract

Large Language Models provide fluent responses but often lack transparency and verifiable evidence. This paper proposes an Explainable AI (XAI) framework that combines GPT-based reasoning, conditional Retrieval-Augmented Generation (RAG), intelligent tool routing, and a Research Agent. Unlike conventional RAG systems that execute retrieval for every query, the proposed framework first analyzes the query to determine whether retrieval is necessary. This reduces latency, computational cost, and unnecessary processing while improving explainability.

Keywords: Explainable AI, RAG, GPT, Research Agent, Tool Routing, Whisper, FFmpeg, Document Intelligence.

1. Introduction

Generative AI systems increasingly support decision making in education, healthcare, finance, and research. However, users often require evidence supporting generated answers. Existing systems may execute retrieval pipelines even when unnecessary, increasing response time and cost. This work introduces a conditional retrieval strategy combined with an explainability layer and a research-focused agent.

2. Proposed Architecture

The workflow begins with a Query Analyzer that classifies user intent and query complexity. Simple knowledge-based questions are answered directly by GPT. Document-based questions invoke the RAG pipeline. Supported inputs include PDF, Word, PowerPoint, Excel, images, audio, and video. Video files are processed using FFmpeg for audio extraction and Whisper for speech-to-text conversion. Extracted content is passed to the retrieval module, whose output is forwarded to GPT for reasoning.

3. Research Agent

After retrieval, a dedicated Research Agent expands the context by identifying approximately five relevant research papers from trusted scholarly sources. It summarizes key findings, compares methodologies, identifies supporting or conflicting evidence, and prepares a concise evidence package for the language model. This enables evidence-backed answers rather than responses based solely on retrieved document chunks.

4. Explainability

The framework records why RAG was executed or skipped, which tools were selected, document provenance, confidence scores, and the reasoning path. These explanations improve transparency, user trust, and auditability while keeping the architecture simple.

5. Advantages

- Reduced latency through conditional RAG.
- Lower computational cost by avoiding unnecessary retrieval.
- Modular support for heterogeneous document formats.
- Evidence-backed responses using a Research Agent.
- Transparent decision making with explainable workflows.

6. Conclusion and Future Work

The proposed XAI framework integrates intelligent query analysis, conditional retrieval, GPT reasoning, and research-driven evidence aggregation into a unified architecture. Future work includes adaptive tool selection, multimodal reasoning, confidence calibration, citation verification, and benchmark evaluation against traditional RAG systems.